

FACTSHEET FS11 · Residual momentum

Momentum without the market's ups and downs

Henrik Gjerner · Rude Investment Consulting · Project 10 sidebar: strategy factsheets · data to 31 August 2026

STRATEGY	ORIGIN	FAMILY
Residual (idiosyncratic) momentum	Blitz, Huij & Martens (2011)	Momentum & trend
UNIVERSES	BACKTEST	BOOKS
US · EU · UK · DK · SCANDI · World (point-in-time)	Feb 2013 – Aug 2026, monthly, net of costs	Long/short and long-only (stocks with the strongest stock-specific trend)
REBALANCE	STATUS	
Monthly	Descriptive factsheet, not a registered trial	

The claim. Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009). **What we find, 2013–2026, point-in-time universes, after costs:** the long/short earned +1.9% to +7.9% a year and was positive in all six; no universe passes the 2.87 gate. The long-only book (stocks with the strongest stock-specific trend) had a higher Sharpe than the equal-weight universe in all six, with alphas of +2.7% to +4.4%, none past the gate. **What goes wrong (§10):** the main drags are a market bet (average beta -0.26, worth -2.8% a year in 2013–19), costs of about 2.0% a year. None of the pre-declared fixes passes the gate in both windows. **Classification (§11):** Momentum & trend; its returns sit closest to the momentum cluster of the signal library. **360° view (§12):** nearest neighbours Momentum 12-1, Above 52-week low; after those neighbours and the market, nothing is left (alpha $|t| < 2$ everywhere): the signal is redundant.

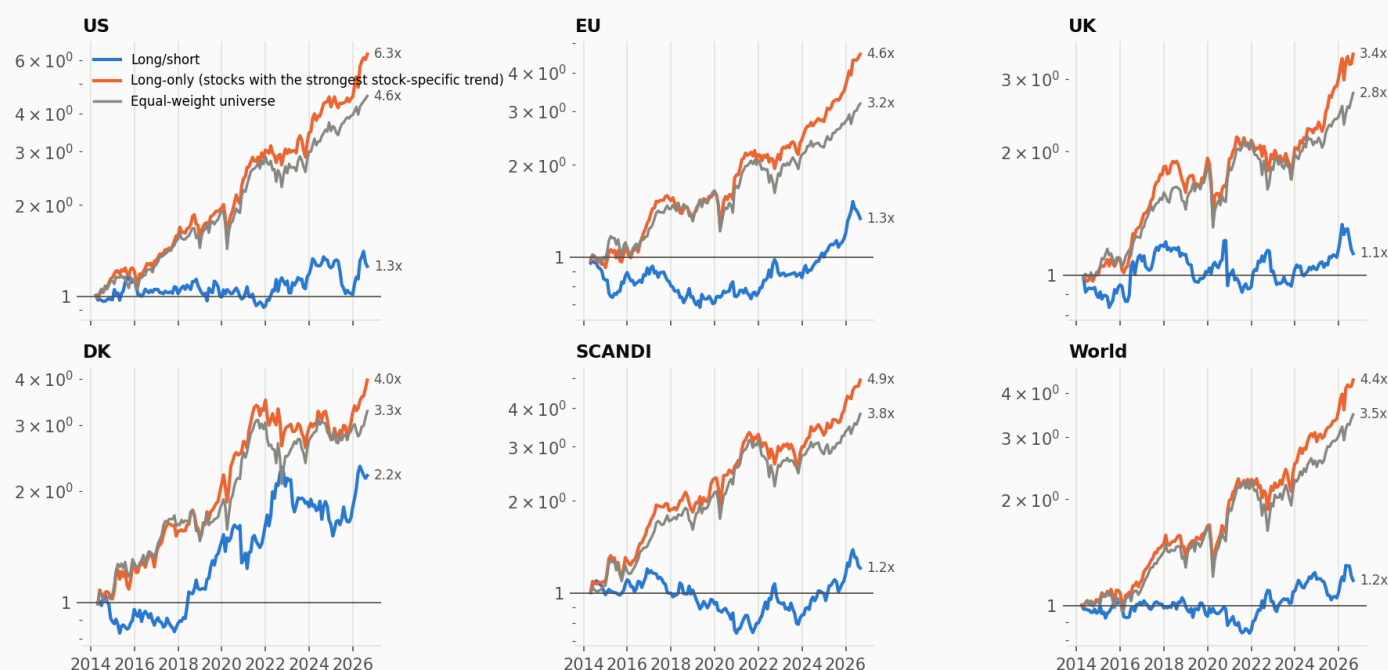
1. Headline performance

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long- only CAGR	Long- only Sharpe	Long- only max DD	EW CAGR	EW Sharpe	Size- weighted CAGR
US	497 / 10	+2.8%	0.79	+5.2% (1.5)	-0.18	+15.8%	0.98	-24%	+12.9%	0.87	+14.5%
EU	238 / 10	+3.1%	0.85	+6.0% (1.6)	-0.28	+13.1%	0.94	-21%	+9.7%	0.70	+9.2%
UK	235 / 10	+2.0%	0.57	+4.6% (1.4)	-0.28	+10.5%	0.75	-28%	+8.6%	0.65	+8.3%
DK	19 / 3	+7.9%	1.64	+10.7% (2.3)	-0.26	+11.7%	0.76	-26%	+10.0%	0.68	+8.0%
SCANDI	62 / 5	+2.4%	0.65	+4.7% (1.2)	-0.20	+13.7%	0.94	-24%	+11.4%	0.82	+9.7%

Universe	Names / groups	L/S return / yr	L/S t	L/S CAPM alpha (t)	L/S beta	Long-only CAGR	Long-only Sharpe	Long-only max DD	EW CAGR	EW Sharpe	Size-weighted CAGR
World	965 / 10	+1.9%	0.68	+4.3% (1.7)	-0.21	+12.5%	0.83	-26%	+10.5%	0.71	+9.7%

Monthly, local currency (World in USD). L/S = equal-weighted stocks with the strongest stock-specific trend minus the weakest, net of 10 bp per side on turnover, a size-tiered borrow fee on the short leg and, where the book is net long cash, financing at the USD risk-free rate + 0.5%. Newey-West t (6 lags). CAPM vs the equal-weight universe. Size-weighted universe = market-cap weights in the US, traded-value weights elsewhere (a proxy for the index).

Residual momentum on six stock universes, 2013-2026



Growth of 1, log scale, local currency, net of costs. Henrik Gjernerig · Project 10 factsheets

2. Strategy description

Ranking stocks on the part of their past return that the market does not explain gives a momentum strategy with about half the volatility and far smaller crashes than plain momentum (Blitz, Huij & Martens 2011, US 1930–2009).

Why it might work. Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).

Rule	Original (Blitz, Huij & Martens (2011))	This factsheet
Signal	Sum of Fama-French 3-factor residuals $t-12..t-2$, over a 36-month window, divided by their volatility	Market-model residuals over 24 months, same scaling
Portfolio	Decile L/S, one month	Extreme quantile L/S, one month
Weighting	Equal weighted	Equal weighted

3. Data load

Universe	Prices	Membership	Names eligible (median)	Currency
US	Sharadar SEP <code>closeadj</code> (total return), 805 tickers incl. delisted	Point-in-time: top 500 by market cap each month-end (Sharadar filings × price, A06 method)	~499	USD
EU	Project1 yfinance cache, Adj Close (TR)	Point-in-time: union of 11 national blue-chip indices (CAC 40, DAX, AEX, IBEX 35, FTSE MIB, OMXS30, OMXC25, OMXH25, BEL 20, PSI-20, WIG20), quarterly Wikipedia snapshots 2012–2026; 83% of member-quarters priced	~238	local (EUR, SEK, DKK, PLN)
UK	Project1 cache, Close + pence dividends (Yahoo under-adjusts LSE ~100×)	Point-in-time: FTSE 100 + FTSE 250, quarterly snapshots; 75% priced	~242	GBP
DK	Project1 cache, Adj Close	Point-in-time: OMX Copenhagen 25, quarterly snapshots; 91% priced	~19	DKK
SCANDI	Project1 cache, Adj Close	Point-in-time: OMXC25 + OMXS30 + OMXH25 (Oslo has no snapshot history); 88% priced	~62	local (DKK, SEK, EUR)
World	US Sharadar + UK + EU above	Point-in-time union of the three; local-currency returns summed, not converted	~963	mixed

Membership comes from Project2's cache of dated Wikipedia index pages (the same parsing as `run_wiki_pit.py` / `run_eu_pit.py`), with a short list of verified ticker renames (e.g. `NZYM-B` → `NSIS-B`, `WDH` → `DEMANT`, `DAI` → `MBG`). A name is eligible from the day after the snapshot that lists it, once it has 60 days of prices. Member-quarters without a price are mostly delisted names that Yahoo no longer serves, so some survivorship remains; see §13.

- **Data gap:** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.

4. Signal creation

$\text{RESMOM}_{i,t} = \Sigma \epsilon_{i,t-11..t-1} / \sigma(\epsilon)$, from monthly regressions of r_i on the equal-weight universe over the last 24 months. Signals use only information up to month-end t and are traded in month $t+1$. Eligibility: in the point-in-time universe on any of the last five trading days of month t .

5. Model build

- Sort eligible stocks into equal-count groups (deciles ≥ 100 names, quintiles 50–99, terciles < 50 ; fixed per universe).

- **Long/short** = stocks with the strongest stock-specific trend minus the weakest; **long-only** = stocks with the strongest stock-specific trend; benchmarks: equal-weight universe and size-weighted universe.
- Equal weights within each leg, rebalanced monthly.
- Costs: 10 bp per side on the one-way turnover of each leg; short-leg borrow fee 0.25% / 0.75% / 2% a year by size tier (largest 50% / next 30% / smallest 20% of the universe, `code/borrow.py`); net long cash financed at the USD risk-free rate + 0.5%. Code: `code/xs.py` (engine), `code/analyse_xs.py`, `code/build_xs.py`.

6. Performance in detail

US (top 500, point-in-time), median 497 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.8%	+1.8%	14.1%	0.20 [-0.27, 0.63]	0.20	-25%	46%	0.79	-0.18	+5.2% (1.5)	8.1%
Long-only (net)	+16.1%	+15.8%	16.5%	0.98 [0.55, 1.46]	1.67	-24%	64%	4.42	0.94	+3.5% (2.0)	9.2%
Stocks with the strongest stock-specific trend (gross)	+17.2%	+17.1%	16.5%	1.05 [0.61, 1.54]	1.83	-24%	66%	4.72	0.94	+4.6% (2.6)	9.1%
The weakest (gross)	+11.4%	+10.0%	19.0%	0.60 [0.17, 1.13]	0.82	-32%	62%	2.78	1.12	-3.6% (-1.9)	11.4%
EW universe	+13.4%	+12.9%	15.5%	0.87 [0.43, 1.42]	1.34	-25%	68%	4.23	1.00	—	9.2%
Size-weighted universe	+14.7%	+14.5%	14.7%	1.00 [0.53, 1.57]	1.64	-24%	69%	4.69	0.90	+2.6% (1.9)	8.6%

EU (11 national blue-chip indices, point-in-time), median 238 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+3.1%	+2.4%	12.3%	0.25 [-0.33, 0.84]	0.29	-29%	55%	0.85	-0.28	+6.0% (1.6)	6.7%
Long-only (net)	+13.3%	+13.1%	14.2%	0.94 [0.44, 1.54]	1.63	-21%	62%	3.48	0.85	+4.4% (2.1)	7.5%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Stocks with the strongest stock-specific trend (gross)	+14.4%	+14.3%	14.1%	1.02 [0.51, 1.62]	1.81	-21%	62%	3.77	0.85	+5.5% (2.6)	7.4%
The weakest (gross)	+8.3%	+6.9%	18.3%	0.45 [-0.01, 0.94]	0.61	-37%	56%	1.82	1.13	-3.5% (-1.6)	10.5%
EW universe	+10.4%	+9.7%	14.8%	0.70 [0.24, 1.24]	1.06	-26%	62%	2.88	1.00	–	8.6%
Size-weighted universe	+9.9%	+9.2%	14.5%	0.68 [0.20, 1.22]	1.01	-26%	58%	2.70	0.95	+0.0% (0.0)	8.6%

UK (FTSE 350, point-in-time), median 235 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.0%	+1.0%	14.0%	0.14 [-0.35, 0.63]	0.10	-25%	54%	0.57	-0.28	+4.6% (1.4)	8.8%
Long-only (net)	+11.1%	+10.5%	14.9%	0.75 [0.22, 1.38]	1.12	-28%	59%	2.89	0.91	+2.7% (1.3)	9.0%
Stocks with the strongest stock-specific trend (gross)	+12.2%	+11.7%	14.9%	0.82 [0.29, 1.46]	1.26	-28%	59%	3.18	0.91	+3.8% (1.8)	8.9%
The weakest (gross)	+7.2%	+5.6%	19.1%	0.38 [-0.07, 0.89]	0.45	-35%	58%	1.57	1.19	-3.8% (-2.0)	11.5%
EW universe	+9.3%	+8.6%	14.4%	0.65 [0.17, 1.23]	0.90	-29%	58%	2.70	1.00	–	9.2%
Size-weighted universe	+8.8%	+8.3%	12.9%	0.69 [0.17, 1.33]	0.97	-27%	62%	2.66	0.82	+1.2% (0.6)	8.0%

DK (OMXC25, point-in-time), median 19 names, 3 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+7.9%	+6.6%	17.3%	0.46 [-0.09, 1.02]	0.57	-35%	55%	1.64	-0.26	+10.7% (2.3)	11.1%

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
Long-only (net)	+12.5%	+11.7%	16.5%	0.76 [0.26, 1.25]	1.18	-26%	60%	3.23	0.88	+2.9% (1.4)	8.7%
Stocks with the strongest stock- specific trend (gross)	+13.2%	+12.5%	16.5%	0.80 [0.30, 1.30]	1.27	-25%	60%	3.40	0.88	+3.6% (1.7)	8.6%
The weakest (gross)	+3.2%	+1.1%	20.8%	0.16 [-0.38, 0.73]	0.07	-55%	56%	0.56	1.14	-9.1% (-3.2)	14.1%
EW universe	+10.9%	+10.0%	15.9%	0.68 [0.17, 1.26]	0.98	-33%	62%	2.62	1.00	–	9.6%
Size-weighted universe	+9.1%	+8.0%	16.8%	0.54 [-0.01, 1.17]	0.70	-40%	59%	2.01	0.91	-0.8% (-0.3)	11.2%

SCANDI (OMXC25 + OMXS30 + OMXH25, point-in-time), median 62 names, 5 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+2.4%	+1.5%	13.1%	0.18 [-0.35, 0.71]	0.18	-38%	48%	0.65	-0.20	+4.7% (1.2)	7.4%
Long-only (net)	+14.0%	+13.7%	14.9%	0.94 [0.42, 1.53]	1.65	-24%	61%	3.75	0.90	+3.3% (1.8)	7.6%
Stocks with the strongest stock- specific trend (gross)	+14.9%	+14.8%	14.9%	1.00 [0.48, 1.60]	1.80	-22%	61%	3.99	0.90	+4.2% (2.3)	7.5%
The weakest (gross)	+10.0%	+8.8%	17.7%	0.57 [0.01, 1.16]	0.76	-46%	61%	2.08	1.10	-3.1% (-1.3)	11.2%
EW universe	+11.9%	+11.4%	14.5%	0.82 [0.29, 1.43]	1.25	-29%	66%	3.14	1.00	–	8.5%
Size-weighted universe	+10.3%	+9.7%	14.2%	0.73 [0.22, 1.30]	1.07	-27%	62%	2.80	0.94	-0.9% (-0.8)	8.9%

World (US + UK + EU, point-in-time), median 965 names, 10 groups

Book	Mean / yr	CAGR	Vol	Sharpe [95% CI]	Sortino	Max DD	Hit	t(mean)	Beta	Alpha/yr (t)	CVaR 95% (month)
L/S (net)	+1.9%	+1.3%	10.8%	0.18 [-0.30, 0.65]	0.18	-21%	48%	0.68	-0.21	+4.3% (1.7)	6.3%
Long-only (net)	+13.1%	+12.5%	15.7%	0.83 [0.37, 1.35]	1.28	-26%	66%	3.66	0.92	+2.7% (2.0)	9.6%
Stocks with the strongest stock- specific trend (gross)	+14.1%	+13.7%	15.7%	0.90 [0.44, 1.43]	1.43	-25%	67%	3.97	0.92	+3.8% (2.8)	9.5%
The weakest (gross)	+9.2%	+7.7%	19.0%	0.49 [0.01, 1.02]	0.63	-40%	60%	2.00	1.13	-3.5% (-2.4)	11.7%
EW universe	+11.3%	+10.5%	16.0%	0.71 [0.25, 1.27]	1.02	-28%	63%	3.03	1.00	–	9.8%
Size-weighted universe	+10.5%	+9.7%	15.2%	0.69 [0.20, 1.26]	1.00	-28%	62%	2.80	0.93	-0.1% (-0.1)	9.3%

Sharpe CI: stationary bootstrap (mean block 6 months, 5,000 draws). Max drawdown on monthly returns.

Calendar-year returns, L/S (net)

Year	US	EU	UK	DK	SCANDI	World
2014	-2.6%	-15.8%	-11.5%	-11.0%	-0.9%	-4.1%
2015	+14.5%	+5.8%	+7.3%	+7.9%	+11.8%	+6.5%
2016	-5.7%	-1.2%	+14.3%	-6.5%	+5.2%	-1.5%
2017	-3.1%	-8.5%	+8.4%	-1.9%	-14.0%	-2.1%
2018	+3.3%	-7.8%	-4.3%	+23.3%	+2.3%	+1.1%
2019	-5.6%	-4.7%	-9.7%	+34.9%	-9.9%	-6.3%
2020	-0.6%	+4.9%	-1.0%	-10.5%	-19.6%	-3.6%
2021	-6.0%	+2.9%	-6.8%	+23.9%	+8.0%	-5.3%
2022	+20.8%	+13.9%	+9.3%	+33.3%	+16.2%	+20.6%
2023	+5.4%	-1.6%	+1.2%	-16.3%	-8.7%	+7.1%
2024	+4.8%	+16.7%	-0.5%	-10.3%	+17.7%	+8.0%
2025	-18.3%	+19.5%	+12.4%	+11.6%	+15.5%	-8.3%
2026	+23.8%	+11.9%	-2.7%	+21.3%	+4.4%	+8.2%

Calendar-year returns, long-only (net)

Year	US	EU	UK	DK	SCANDI	World
2014	+15.3%	-7.7%	+0.4%	+2.5%	+8.6%	+3.5%
2015	+5.1%	+12.9%	+9.0%	+23.0%	+17.7%	+5.1%
2016	+8.5%	+26.6%	+24.9%	+7.1%	+30.3%	+8.9%
2017	+19.6%	+15.7%	+34.6%	+15.5%	+12.5%	+26.6%
2018	-2.6%	-13.1%	-13.5%	-3.0%	-1.8%	-10.9%
2019	+28.0%	+21.6%	+21.1%	+38.1%	+28.8%	+25.9%
2020	+24.8%	+14.9%	-0.3%	+30.2%	+11.1%	+18.2%
2021	+24.1%	+16.5%	+10.9%	+29.1%	+25.1%	+14.5%
2022	-2.7%	-3.8%	-9.8%	-15.1%	-11.6%	-6.8%
2023	+17.1%	+16.6%	+6.5%	-0.9%	+3.0%	+19.2%
2024	+21.8%	+14.3%	+11.0%	-6.1%	+9.1%	+15.2%
2025	+7.9%	+31.6%	+33.0%	+17.4%	+27.9%	+22.4%
2026	+38.2%	+26.2%	+14.1%	+22.1%	+18.0%	+22.0%

7. Trading record

The rebalance log per universe is in `results/xs_FS11_record_<universe>.csv` (names per leg, turnover, leg returns, gross and net L/S).

Universe	Rebalances	Names long / short	Turnover long leg / month	Turnover short leg / month	L/S months positive	Worst L/S month	When
US	150	50 / 49	46%	47%	46%	-9.9%	2025-06
EU	149	24 / 23	44%	46%	55%	-10.3%	2020-11
UK	149	25 / 24	45%	46%	54%	-22.3%	2020-11
DK	149	7 / 7	28%	28%	55%	-20.7%	2020-11
SCANDI	149	13 / 12	38%	38%	48%	-11.9%	2020-11
World	150	98 / 97	46%	47%	48%	-10.1%	2020-11

Current book (signal at 31 August 2026; first 8 names per leg)

Universe	Long: stocks with the strongest stock-specific trend	Short: the weakest
US	PFG, HST, FRT, TGT, EIX, IEX, CSX, BIIB	VST, UBER, PLTR, CSGP, RDDT, NFLX, META, BSX

Universe	Long: stocks with the strongest stock-specific trend	Short: the weakest
EU	DANSKE.CO, INVE-B.ST, DHL.DE, MRK.DE, SWED-A.ST, INDU-C.ST, NESTE.HE, LOG.MC	BAR.BR, CTT.LS, AGFB.BR, IVG.MI, RMS.PA, G24.DE, CBK.DE, RHM.DE
UK	KLR.L, EMG.L, MRCH.L, BLND.L, RTW.L, ROSE.L, CRDA.L, LAND.L	RNK.L, ENT.L, STJ.L, RMV.L, HWG.L, AVON.L, FCSS.L, AGT.L
DK	DANSKE.CO, NKT.CO, NOVO-B.CO, CARL-B.CO, JYSK.CO, ORSTED.CO, RBREW.CO, NDA-DK.CO	BAVA.CO, TRYG.CO, MAERSK-A.CO, MAERSK-B.CO, COLO-B.CO, NSIS-B.CO, DSV.CO, AMBU-B.CO
SCANDI	DANSKE.CO, INVE-B.ST, SWED-A.ST, NESTE.HE, INDU-C.ST, ATCO-A.ST, TYRES.HE, NOVO-B.CO	BAVA.CO, KEMIRA.HE, MANTA.HE, SKF-B.ST, KNEBV.HE, VALMT.HE, ORNBV.HE, TEL2-B.ST

8. Risk and factor attribution

JKP 7 themes (US, UK, DK, World) or French Europe 5F + WML (EU, SCANDI); Newey-West t, 6 lags.

Long/short

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| US | JKP USA 7 themes | -0.6% | -0.21 | 0.51 | size +0.31 (+2.7), value -0.31 (-2.5), momentum +0.92 (+8.7), low_risk +0.54 (+3.0), quality -0.68 (-2.3) || EU | French Europe 5F + WML | -4.2% | -1.40 | 0.43 | HML +0.42 (+2.6), WML +0.78 (+9.9) || UK | JKP GBR 7 themes | +0.3% | 0.09 | 0.51 | momentum +0.97 (+5.9) || DK | JKP DNK 7 themes | +1.0% | 0.26 | 0.32 | momentum +0.86 (+5.5), short_term_reversal +0.44 (+2.3) || SCANDI | French Europe 5F + WML | -2.6% | -0.78 | 0.18 | WML +0.51 (+5.2) || World | JKP World 7 themes | +1.1% | 0.37 | 0.42 | value -0.31 (-2.3), momentum +0.67 (+7.6), low_risk +0.35 (+2.5), quality -0.72 (-3.6) |

Long-only

| Universe | Factor model | Alpha / yr | t(alpha) | R² | Loadings with |t| ≥ 2 (t) | ---|---|---|---|---|---| US | JKP USA 7 themes | +4.1% | 2.80 | 0.87 | mkt +1.05 (+24.7), momentum +0.45 (+6.0), low_risk +0.32 (+2.3), quality -0.34 (-2.2) || EU | French Europe 5F + WML | +6.5% | 2.50 | 0.64 | Mkt-RF +0.71 (+14.0), HML +0.35 (+2.9) || UK | JKP GBR 7 themes | +4.7% | 2.15 | 0.71 | mkt +0.59 (+8.9), momentum +0.47 (+6.2), low_risk -0.56 (-2.9) || DK | JKP DNK 7 themes | +4.5% | 1.65 | 0.58 | mkt +0.67 (+8.5), momentum +0.25 (+2.1), low_risk -0.31 (-2.5), short_term_reversal +0.30 (+2.3) || SCANDI | French Europe 5F + WML | +8.0% | 2.28 | 0.49 | Mkt-RF +0.65 (+8.6) || World | JKP World 7 themes | +7.2% | 3.35 | 0.79 | mkt +0.74 (+13.3), momentum +0.30 (+3.1) |

9. Statistical verdict

- **Gate:** |t| > 2.87 (2 books × 6 universes, Bonferroni 0.05/12, two-sided).
- **L/S mean:** positive in all six; passes in 0 of 6.
- **L/S CAPM alpha:** +4.3% to +10.7%, |t| past the gate in 0 of 6.
- **Long-only:** Sharpe above the equal-weight universe in all six; alpha +2.7% to +4.4%, passing in 0 of 6.
- **Probabilistic Sharpe** of the L/S, P(true Sharpe > 0): US 0.76, EU 0.81, UK 0.69, DK 0.94, SCANDI 0.74, World 0.73.
- **Publication decay:** gross L/S -2.3% to +9.6% a year in 2013–19 against +5.7% to +13.3% in 2020–26.

10. What goes wrong, and how to fix it

Method. Diagnose on the design window (2013–2019); apply the same pre-declared fix ladder used for every sort strategy in FS03–FS12; judge on the holdout (Jan 2020 – Aug 2026) with a pooled six-universe paired bootstrap of the Sharpe difference against the previous step. 25 fix trials across FS03–FS12, gate $p < 0.0020$.

10.1 Diagnosis (design window 2013–2019)

Universe	L/S gross / yr	Beta	Beta drag / yr	CAPM alpha (gross)	Costs / yr	Long leg vs EW / yr	EW vs short leg / yr	Turnover / month	Worst months
US	+3.7%	-0.35	-4.1%	+7.8%	-2.2%	+1.5%	+2.1%	93%	2016-02 -8%, 2019-11 -7%, 2015-10 -7%
EU	-2.3%	-0.37	-3.5%	+1.2%	-2.2%	+0.6%	-2.9%	91%	2015-02 -9%, 2018-11 -6%, 2014-11 -6%
UK	+4.1%	-0.36	-4.0%	+8.1%	-2.2%	+1.9%	+2.3%	93%	2014-05 -8%, 2016-08 -7%, 2019-04 -6%
DK	+9.6%	-0.05	-0.6%	+10.2%	-1.4%	+3.1%	+6.5%	58%	2014-11 -8%, 2016-04 -7%, 2015-04 -5%
SCANDI	+1.7%	-0.09	-1.2%	+2.9%	-1.8%	+2.8%	-1.1%	73%	2018-07 -6%, 2017-10 -6%, 2018-11 -6%
World	+2.3%	-0.33	-3.1%	+5.4%	-2.3%	+1.1%	+1.2%	95%	2016-02 -7%, 2019-04 -6%, 2015-02 -5%

On average across the six universes the gross spread was +3.2% a year, of which the market exposure (beta -0.26) contributed -2.8%; the beta-adjusted spread (CAPM alpha) was +5.9%. Costs took 2.0% a year at 84% monthly turnover across both legs. The long leg beat the universe by +1.9% and the short leg lagged it by +1.3% a year, so most of the spread comes from the long side. Holding longer does not change the picture much: the gross spread 2, 3 and 6 months after formation averages +6.6%, +4.0%, +4.0% a year.

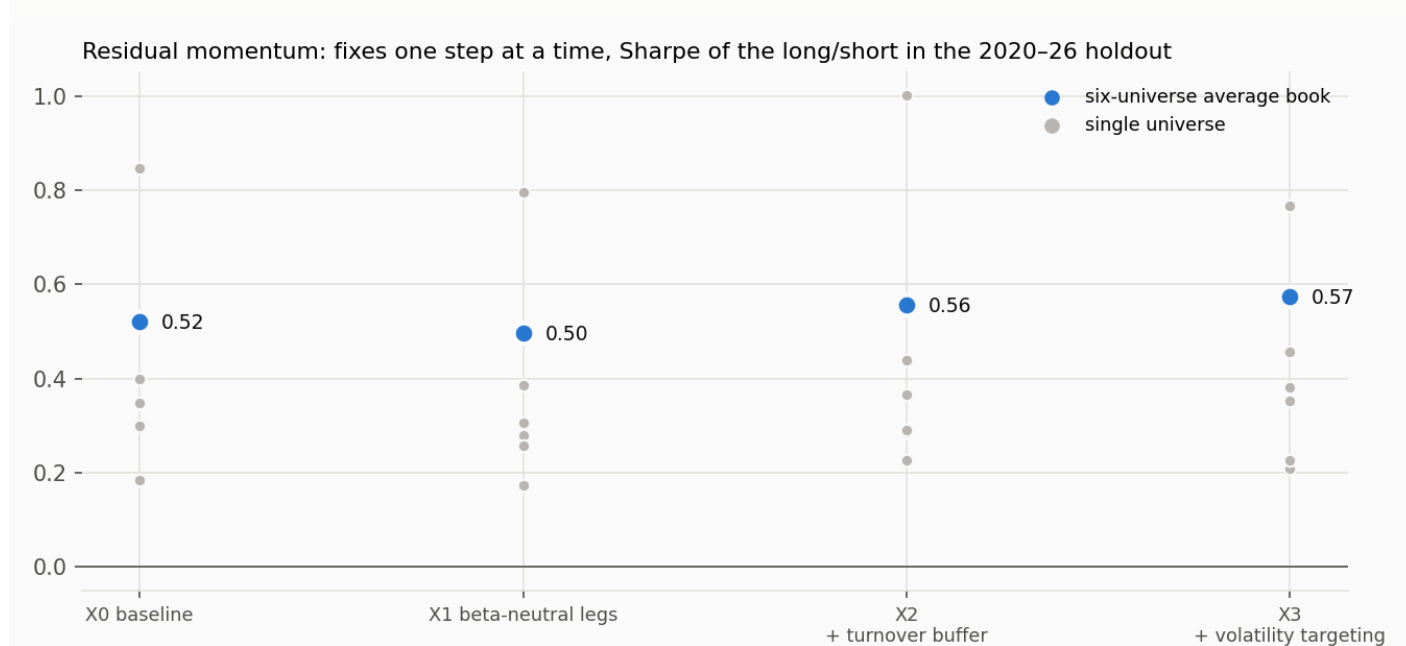
10.2 The fix ladder

X0 baseline → X1 **beta-neutral legs** (each leg scaled by 1/its ex-ante beta) → X2 + **turnover buffer** (enter the extreme 1/q, keep a name until it leaves the extreme 3/q) → X3 + **volatility targeting** (scale the L/S to 10% a year using its trailing 6-month volatility, lagged; Barroso & Santa-Clara 2015; Moreira & Muir 2017).

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X0 baseline	0.02	–	0.52	–	–	+5.9%	-15%	+9.0% (2.3)	2.0%

Step	Design Sharpe (avg)	Δ design (p)	Holdout Sharpe (book)	Δ holdout (p)	Universes improved	Holdout return / yr	Holdout max DD	Holdout alpha vs EW (t)	Costs / yr
X1 beta-neutral legs	0.06	+0.08 (p 0.224)	0.50	-0.02 (p 0.567)	2 of 6	+4.7%	-13%	+6.2% (1.6)	2.1%
X2 + turnover buffer	0.08	+0.05 (p 0.388)	0.56	+0.06 (p 0.272)	5 of 6	+4.6%	-13%	+5.7% (1.7)	1.1%
X3 + volatility targeting	0.04	-0.09 (p 0.576)	0.57	+0.02 (p 0.415)	3 of 6	+5.5%	-16%	+6.5% (1.9)	1.0%

✓ = passes the gate in the holdout. Design Sharpe = average across universes; tests and holdout columns use the six-universe equal-weighted book.



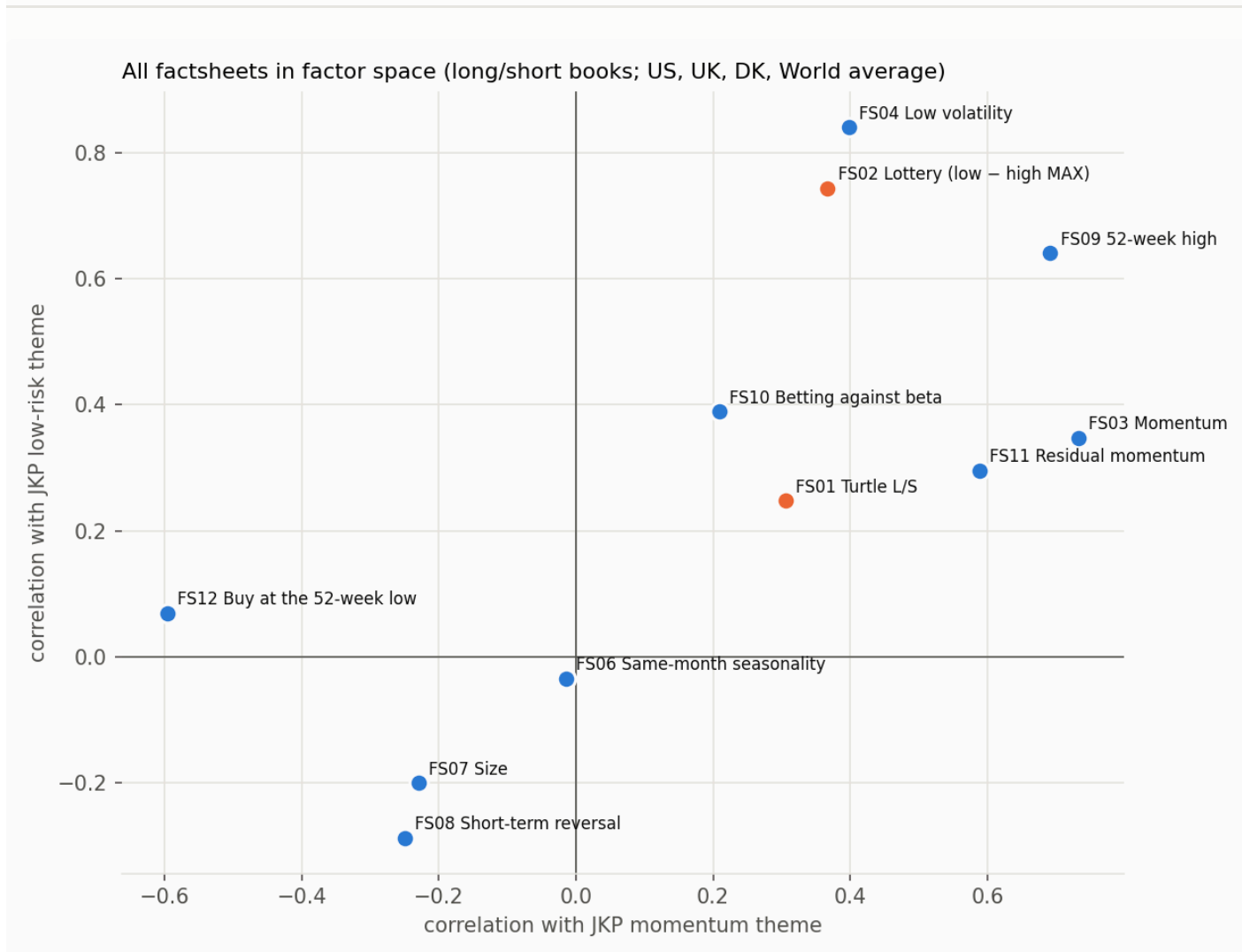
- **X1 beta-neutral legs:** +0.08 design, -0.02 holdout; helps in one window and hurts in the other: not reliable.
- **X2 + turnover buffer:** +0.05 design, +0.06 holdout; helps in both windows but does not pass the gate.
- **X3 + volatility targeting:** -0.09 design, +0.02 holdout; helps in one window and hurts in the other: not reliable.

Where it ends: holdout Sharpe 0.52 → 0.57, return +5.9% → +5.5% a year (six-universe book).

11. Classification: where does it fit?

Dimension	Residual momentum
Series family	Momentum & trend
Academic style	Residual (idiosyncratic) momentum
Signal	Price only (daily total returns)

Dimension	Residual momentum
Cross-section vs time series	Cross-sectional: stocks ranked against each other each month
Horizon and turnover	One month; 84% of the two legs replaced monthly
Market exposure	L/S beta -0.28 to -0.18
Payoff shape	Treynor–Mazuy γ -2.00 to 0.06 (t -3.3 to 0.1); worst 10% of market months +0.5% to +3.1% a month, best 10% -2.0% to -1.2%
Nearest factor theme	momentum (4 of 6 universes), WML (2 of 6 universes)
Nearest library signals (returns)	Momentum 12-1 (6 universes), Above 52-week low (6 universes), Near 52-week high (6 universes)
Economic rationale	Same under-reaction story as momentum, but removing the market (and factor) component avoids the time-varying beta that makes plain momentum crash (Grundy & Martin 2001; Daniel & Moskowitz 2016).
Publication	Grundy & Martin 2001; Gutierrez & Pirinsky 2007; Blitz, Huij & Martens 2011
Where it fits	In the momentum cluster of the library; fully explained by its neighbours.

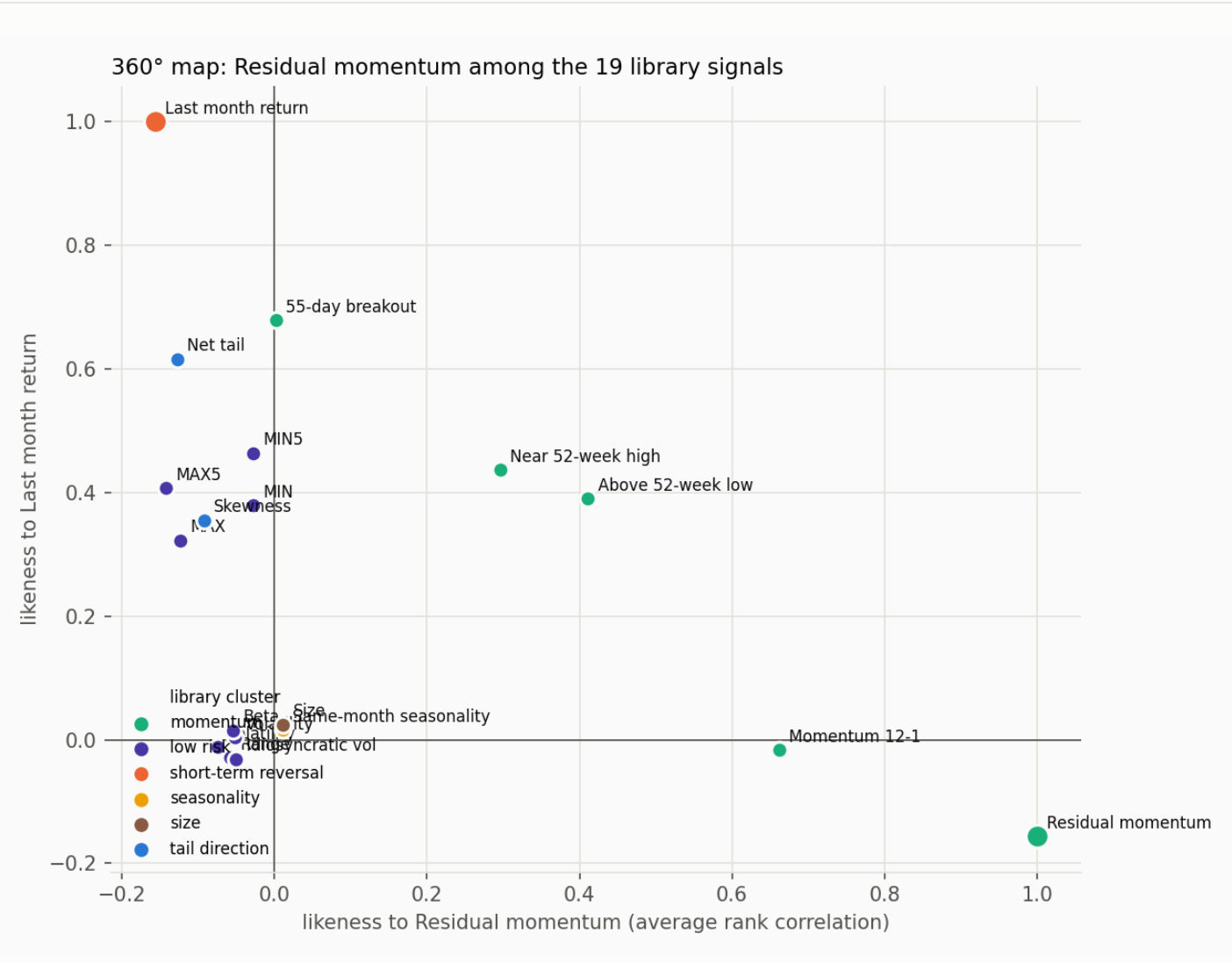


12. 360° view

The signal among the 19 price signals of the shared library (`code/library.py`), on the same universes and months (equal-weighted top minus bottom quantile, gross). Its natural mirror here is **Last month return**.

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Residual momentum (this strategy)	momentum	+1.00	+1.00	+6.1%	+7.3% to +13.0%	5 / 0
Momentum 12-1	momentum	+0.66	+0.63	+7.7%	+3.9% to +17.8%	5 / 0
Near 52-week high	momentum	+0.30	+0.47	+2.5%	+8.3% to +18.8%	6 / 0
Above 52-week low	momentum	+0.41	+0.46	+8.8%	+4.4% to +13.8%	4 / 0
Volatility	low risk	-0.07	-0.28	+3.0%	-10.3% to -3.7%	0 / 5
MAX5	low risk	-0.14	-0.26	+2.6%	-8.5% to +0.4%	0 / 2
Range (MAX–MIN)	low risk	-0.06	-0.26	+1.2%	-11.3% to -2.3%	0 / 4
55-day breakout	momentum	+0.00	+0.26	+0.7%	+5.6% to +12.6%	3 / 0
Beta	low risk	-0.05	-0.25	+5.5%	-15.7% to -2.6%	0 / 3
MAX (best day)	low risk	-0.12	-0.25	+1.9%	-9.3% to +1.0%	0 / 2
MIN5	low risk	-0.03	+0.23	-2.0%	+2.4% to +13.2%	4 / 0
Volatility (252 days)	low risk	-0.05	-0.23	+5.1%	-10.3% to -0.2%	0 / 1
Idiosyncratic vol	low risk	-0.05	-0.20	-1.5%	-10.4% to -0.8%	0 / 3
MIN (worst day)	low risk	-0.03	+0.20	-1.5%	+2.9% to +11.0%	3 / 0
Skewness	tail direction	-0.09	-0.12	-0.2%	-5.4% to +4.5%	0 / 1
Size	size	+0.01	+0.11	-1.5%	-3.8% to +4.6%	0 / 0
Net tail (MAX+MIN)	tail direction	-0.13	-0.07	-1.0%	-5.3% to +4.2%	0 / 0
Last month return (mirror)	short-term reversal	-0.16	+0.05	+0.9%	-3.2% to +8.4%	0 / 0

Signal (top minus bottom, raw sign)	Cluster	Likeness: stocks	Likeness: returns	Raw L/S, avg of 6	CAPM alpha range	Universes alpha t $\geq +2$ / ≤ -2
Same-month seasonality	seasonality	+0.01	-0.04	-1.8%	-8.5% to +7.2%	0 / 0



Spanning. Regressing the L/S on its five closest library neighbours (by return correlation) and the market:

| Universe | Alpha after neighbours / yr | t | R² | Neighbours with |t| ≥ 2 | ---|---|---|---|---| | US | +3.8% | 1.43 | 0.40 | Momentum 12-1 +0.31 (+2.3) | | EU | +2.0% | 0.61 | 0.43 | Momentum 12-1 +0.36 (+3.6), MAX -0.25 (-2.6) | | UK | +2.9% | 0.86 | 0.47 | Above 52-week low +0.32 (+2.6) | | DK | +4.9% | 1.45 | 0.48 | Momentum 12-1 +0.77 (+7.1) | | SCANDI | +5.4% | 1.75 | 0.38 | Momentum 12-1 +0.47 (+4.9) | | World | -0.1% | -0.02 | 0.55 | Momentum 12-1 +0.28 (+2.9), Above 52-week low +0.34 (+3.1), 55-day breakout -0.34 (-3.2), Volatility -0.30 (-4.7) |

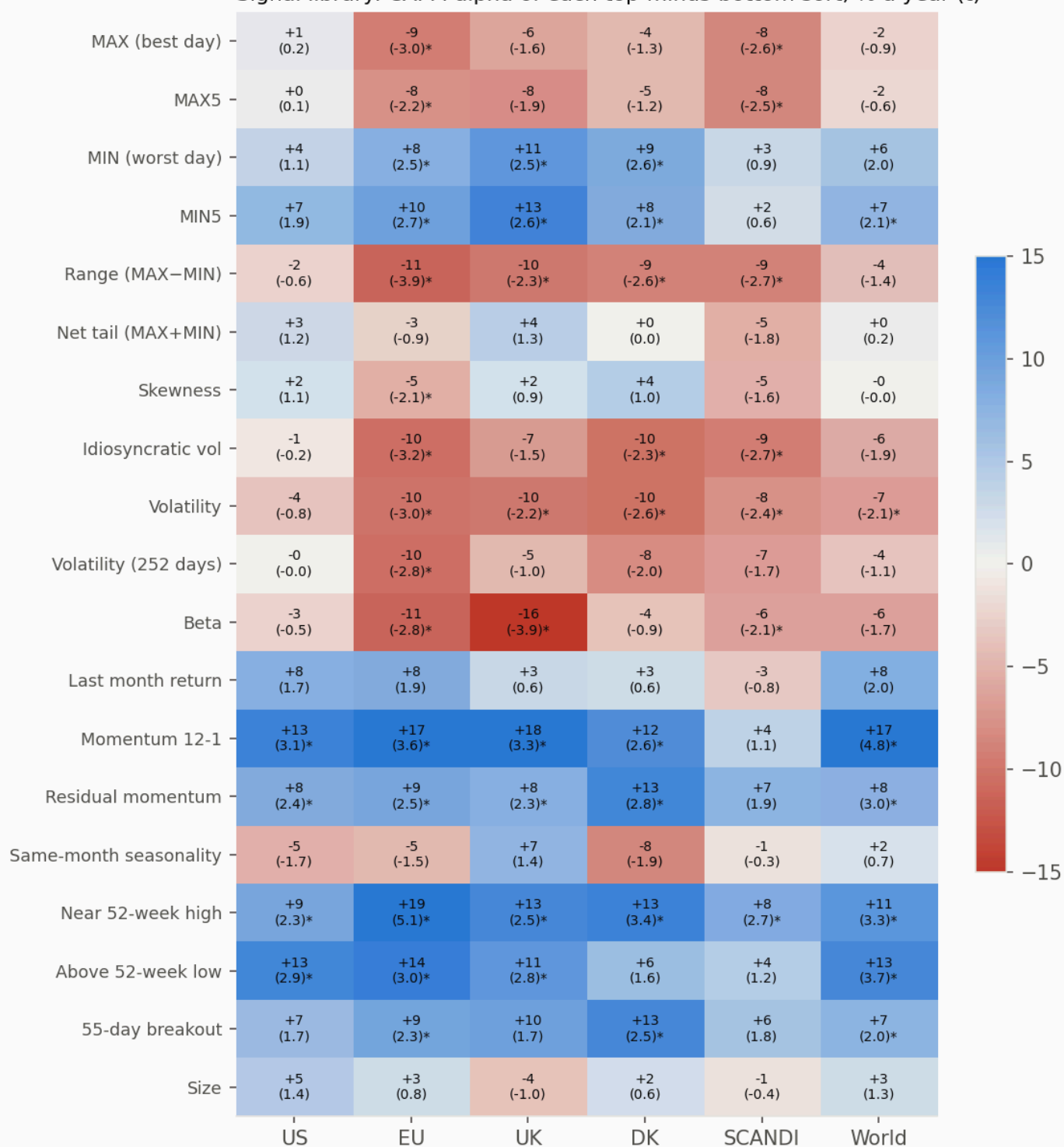
3×3 sort, next-month return a year
(US, EU, UK, SCANDI, World averaged)

low Residual momentum	9.2%	9.0%	10.3%
middle	11.7%	11.4%	11.1%
high Residual momentum	12.9%	13.1%	12.8%
	low Last month return	middle	high Last month return

Holding Last month return fixed (down a column), moving from low to high Residual momentum changes the return by +3.7, +4.1 and +2.5 points a year; holding Residual momentum fixed (along a row), moving from low to high Last month return changes it by +1.1, -0.6 and -0.1. **Market direction.** Top-minus-bottom return in rising vs falling market months (six-universe average): -1% vs +18% a year. **Horizon:** the gross spread at 1, 2, 3 and 6 months after formation averages +6.1%, +6.6%, +4.0%, +4.0% a year.

The full library, with the CAPM alpha of every signal in every universe:

Signal library: CAPM alpha of each top-minus-bottom sort, % a year (t)



13. Caveats

1. **Residual survivorship.** Point-in-time membership everywhere, but 9–25% of member-quarters outside the US have no price (mostly delisted names); long books are flattered and short books penalised by an unknown amount.
2. **Currency.** Local currency; World sums local-currency returns without conversion.
3. **Equal weighting and flat trading costs.** 10 bp per side for every stock and a size-tiered borrow fee, but no market impact; blue-chip universes only.

4. **Data gap.** Market-only residuals over 24 months (the paper uses Fama-French 3 factors over 36); stock-level factor loadings outside the US would need data we do not have. The first signals appear in early 2014.
5. **Not a registered trial.** Descriptive; the fix ladder is counted inside FS03–FS12 (25 trials).

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15. Reproduce

`code/data.py` → `signals.py`, `signals2.py`, `library.py` → `xs.py` + `analyse_xs.py` FS11 → `build_xs.py`
FS11. Metrics use Project1 InvestmentLibrary. Licensed and cached prices are not included.